Configure StorageGRID manually

StorageGRID 11.7

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Configure StorageGRID manually

Create a high availability (HA) group for FabricPool

When configuring StorageGRID for use with FabricPool, you can optionally create one or more high availability (HA) groups. An HA group is a collection of nodes that each contain the StorageGRID Load Balancer service. An HA group can contain Gateway Nodes, Admin Nodes, or both.

You can use an HA group to help keep FabricPool data connections available. An HA group uses virtual IP addresses (VIPs) to provide highly available access to the Load Balancer service. If the active interface in the HA group fails, a backup interface can manage the workload with little impact to FabricPool operations.

For details about this task, see Manage high availability groups. To use the FabricPool setup wizard to complete this task, go to Access and complete the FabricPool setup wizard.

Before you begin

- You have reviewed the best practices for high availability groups.
- You are signed in to the Grid Manager using a supported web browser.
- You have the Root access permission.
- If you plan to use a VLAN, you have created the VLAN interface. See Configure VLAN interfaces.

Steps

1. Select **CONFIGURATION** > **Network** > **High availability groups**.
2. Select **Create**.
3. For the **Enter details** step, complete the following fields.

Field	Description
HA group name	A unique display name for this HA group.
Description (optional)	The description of this HA group.

4. For the **Add interfaces** step, select the node interfaces you want to use in this HA group.

   Use the column headers to sort the rows, or enter a search term to locate interfaces more quickly.

   You can select one or more nodes, but you can select only one interface for each node.

5. For the **Prioritize interfaces** step, determine the Primary interface and any backup interfaces for this HA group.

   Drag rows to change the values in the **Priority order** column.

   The first interface in the list is the Primary interface. The Primary interface is the active interface unless a failure occurs.

   If the HA group includes more than one interface and the active interface fails, the virtual IP (VIP) addresses move to the first backup interface in the priority order. If that interface fails, the VIP addresses

move to the next backup interface, and so on. When failures are resolved, the VIP addresses move back to highest priority interface available.

6. For the **Enter IP addresses** step, complete the following fields.

Field	Description
Subnet CIDR	The address of the VIP subnet in CIDR notation—an IPv4 address followed by a slash and the subnet length (0-32). The network address must not have any host bits set. For example, `192.16.0.0/22`.
Gateway IP address (optional)	Optional. If the ONTAP IP addresses used to access StorageGRID aren't on the same subnet as the StorageGRID VIP addresses, enter the StorageGRID VIP local gateway IP address. The local gateway IP address must be within the VIP subnet.
Virtual IP address	Enter at least one and no more than ten VIP addresses for the active interface in the HA group. All VIP addresses must be within the VIP subnet. At least one address must be IPv4. Optionally, you can specify additional IPv4 and IPv6 addresses.

7. Select **Create HA group** and then select **Finish**.

Create a load balancer endpoint for FabricPool

StorageGRID uses a load balancer to manage the workload from client applications, such as FabricPool. Load balancing maximizes speed and connection capacity across multiple Storage Nodes.

When configuring StorageGRID for use with FabricPool, you must configure a load balancer endpoint and upload or generate a load balancer endpoint certificate, which is used to secure the connection between ONTAP and StorageGRID.

To use the FabricPool setup wizard to complete this task, go to Access and complete the FabricPool setup wizard.

Before you begin

- You are signed in to the Grid Manager using a supported web browser.
- You have the Root access permission.
- You have reviewed the general considerations for load balancing as well as the best practices for load balancing for FabricPool.

Steps

1. Select **CONFIGURATION** > **Network** > **Load balancer endpoints**.
2. Select **Create**.

3. For the **Enter endpoint details** step, complete the following fields.

Field	Description
Name	A descriptive name for the endpoint.
Port	The StorageGRID port you want to use for load balancing. This field defaults to 10433 for the first endpoint you create, but you can enter any unused external port. If you enter 80 or 443, the endpoint is configured only on Gateway Nodes. These ports are reserved on Admin Nodes. **Note:** Ports used by other grid services aren't permitted. See the Network port reference. You will provide this number to ONTAP when you attach StorageGRID as a FabricPool cloud tier.
Client type	Select **S3**.
Network protocol	Select **HTTPS**. **Note**: Communicating with StorageGRID without TLS encryption is supported but not recommended.

4. For the **Select binding mode** step, specify the binding mode. The binding mode controls how the endpoint is accessed—using any IP address or using specific IP addresses and network interfaces.

Option	Description
Global (default)	Clients can access the endpoint using the IP address of any Gateway Node or Admin Node, the virtual IP (VIP) address of any HA group on any network, or a corresponding FQDN. Use the **Global** setting (default) unless you need to restrict the accessibility of this endpoint.
Virtual IPs of HA groups	Clients must use a virtual IP address (or corresponding FQDN) of an HA group to access this endpoint. Endpoints with this binding mode can all use the same port number, as long as the HA groups you select for the endpoints don't overlap.
Node interfaces	Clients must use the IP addresses (or corresponding FQDNs) of selected node interfaces to access this endpoint.
Node type	Based on the type of node you select, clients must use either the IP address (or corresponding FQDN) of any Admin Node or the IP address (or corresponding FQDN) of any Gateway Node to access this endpoint.

5. For the **Tenant access** step, select one of the following:

Field	Description
Allow all tenants (default)	All tenant accounts can use this endpoint to access their buckets. **Allow all tenants** is almost always the appropriate option for the load balancer endpoint used for FabricPool. You must select this option if you have not yet created any tenant accounts.
Allow selected tenants	Only the selected tenant accounts can use this endpoint to access their buckets.
Block selected tenants	The selected tenant accounts can't use this endpoint to access their buckets. All other tenants can use this endpoint.

6. For the **Attach certificate** step, select one of the following:

Field	Description
Upload certificate (recommended)	Use this option to upload a CA-signed server certificate, certificate private key, and optional CA bundle.
Generate certificate	Use this option to generate a self-signed certificate. See Configure load balancer endpoints for details of what to enter.
Use StorageGRID S3 and Swift certificate	This option is available only if you have already uploaded or generated a custom version of the StorageGRID global certificate. See Configure S3 and Swift API certificates for details.

7. Select **Create**.

Changes to an endpoint certificate can take up to 15 minutes to be applied to all nodes.

Create a tenant account for FabricPool

You must create a tenant account in the Grid Manager for FabricPool use.

Tenant accounts allow client applications to store and retrieve objects on StorageGRID. Each tenant account has its own account ID, authorized groups and users, buckets, and objects.

For details about this task, see Create tenant account. To use the FabricPool setup wizard to complete this task, go to Access and complete the FabricPool setup wizard.

Before you begin

- You are signed in to the Grid Manager using a supported web browser.
- You have specific access permissions.

Steps

1. Select **TENANTS**.
2. Select **Create**.
3. For the Enter details steps, enter the following information.

Field	Description
Name	A name for the tenant account. Tenant names don't need to be unique. When the tenant account is created, it receives a unique, numeric account ID.
Description (optional)	A description to help identify the tenant.
Client type	Must be **S3** for FabricPool.
Storage quota (optional)	Leave this field blank for FabricPool.

4. For the Select permissions step:
   a. Don't select **Allow platform services**.

      FabricPool tenants don't typically need to use platform services, such as CloudMirror replication.

   b. Optionally, select **Use own identity source**.
   c. Don't select **Allow S3 Select**.

      FabricPool tenants don't typically need to use S3 Select.

   d. Optionally, select **Use grid federation connection** to allow the tenant to use a grid federation connection for account clone and cross-grid replication. Then, select the grid federation connection to use.
5. For the Define root access step, specify which user will have the initial Root access permission for the tenant account, based on whether your StorageGRID system uses identity federation, single sign-on (SSO), or both.

Option	Do this
If identity federation is not enabled	Specify the password to use when signing into the tenant as the local root user.
If identity federation is enabled	1. Select an existing federated group to have Root access permission for the tenant. 2. Optionally, specify the password to use when signing in to the tenant as the local root user.
If both identity federation and single sign-on (SSO) are enabled	Select an existing federated group to have Root access permission for the tenant. No local users can sign in.

6. Select **Create tenant**.

Create an S3 bucket and obtain access keys

Before using StorageGRID with a FabricPool workload, you must create an S3 bucket for your FabricPool data. You also need to obtain an access key and secret access key for the tenant account you will use for FabricPool.

For details about this task, see Create S3 bucket and Create your own S3 access keys. To use the FabricPool setup wizard to complete this task, go to Access and complete the FabricPool setup wizard.

Before you begin

- You have created a tenant account for FabricPool use.
- You have Root access to the tenant account.

Steps

1. Sign in to the Tenant Manager.

   You can do either of the following:

   - From the Tenant Accounts page in the Grid Manager, select the **Sign in** link for the tenant, and enter your credentials.
   - Enter the URL for the tenant account in a web browser, and enter your credentials.

2. Create an S3 bucket for FabricPool data.

   You must create a unique bucket for each ONTAP cluster you plan to use.

   a. Select **View buckets** from the dashboard, or select **STORAGE (S3)** > **Buckets**.
   b. Select **Create bucket**.
   c. Enter the name of the StorageGRID bucket you want to use with FabricPool. For example, `fabricpool-bucket`.

      You can't change the bucket name after creating the bucket.

   d. Select the region for this bucket.

      By default, all buckets are created in the `us-east-1` region.

   e. Select **Continue**.
   f. Select **Create bucket**.

      Don't select **Enable object versioning** for the FabricPool bucket. Similarly, don't edit a FabricPool bucket to use **Available** or a non-default consistency level. The recommended bucket consistency level for FabricPool buckets is **Read-after-new-write**, which is the default setting for a new bucket.

3. Create an access key and a secret access key.
   a. Select **STORAGE (S3)** > **My access keys**.
   b. Select **Create key**.
   c. Select **Create access key**.

d. Copy the access key ID and the secret access key to a safe location, or select **Download .csv** to save a spreadsheet file containing the access key ID and secret access key.

   You will enter these values in ONTAP when you configure StorageGRID as a FabricPool cloud tier.

   > If you generate a new access key and secret access key in StorageGRID in the future, enter the new keys into ONTAP before deleting the old values from StorageGRID. Otherwise, ONTAP might temporarily lose its access to StorageGRID.

Configure ILM for FabricPool data

You can use this simple example policy as a starting point for your own ILM rules and policy.

This example assumes you are designing the ILM rules and an ILM policy for a StorageGRID system that has four Storage Nodes at a single data center in Denver, Colorado. The FabricPool data in this example uses a bucket named `fabricpool-bucket`.

> The following ILM rules and policy are only examples. There are many ways to configure ILM rules. Before activating a new policy, simulate the proposed policy to confirm it will work as intended to protect content from loss. To learn more, see Manage objects with ILM.

> To avoid data loss, do not use an ILM rule that will expire or delete FabricPool cloud tier data. Set the retention period to **forever** to ensure that FabricPool objects aren't deleted by StorageGRID ILM.

Before you begin

- You have reviewed the best practices for using ILM with FabricPool data.
- You are signed in to the Grid Manager using a supported web browser.
- You have the ILM or Root access permission.
- If you upgraded to StorageGRID 11.7 from a previous StorageGRID version, you have configured the storage pool you will use. In general, you should create a storage pool for each StorageGRID site.

  > This prerequisite does not apply to new StorageGRID 11.7 installations. When you install StorageGRID 11.7 on a new grid, storage pools are automatically created for each site.

Steps

1. Create an ILM rule that applies only to the data in `fabricpool-bucket`. This example rule creates erasure-coded copies.

Rule definition	Example value
Rule name	2 + 1 erasure coding for FabricPool data
Bucket name	`fabricpool-bucket` You could also filter on the FabricPool tenant account.

Rule definition	Example value
Advanced filters	Object size greater than 0.2 MB. **Note:** FabricPool only writes 4 MB objects, but you must add an Object size filter because this rule uses erasure coding.
Reference time	Ingest time
Time period and placements	From Day 0 store forever Store objects by erasure coding using 2+1 EC scheme at Denver and retain those objects in StorageGRID forever. To avoid data loss, do not use an ILM rule that will expire or delete FabricPool cloud tier data.
Ingest behavior	Balanced

2. Create a default ILM rule that will create two replicated copies of any objects not matched by the first rule. Don't select a basic filter (tenant account or bucket name) or any advanced filters.

Rule definition	Example value
Rule name	Two replicated copies
Bucket name	*none*
Advanced filters	*none*
Reference time	Ingest time
Time period and placements	From Day 0 store forever Store objects by replicating 2 copies at Denver.
Ingest behavior	Balanced

3. Create a proposed ILM policy and select the two rules. Because the replication rule does not use any filters, it can be the default (last) rule for the policy.
4. Ingest test objects into the grid.
5. Simulate the policy with the test objects to verify the behavior.
6. Activate the policy.

When this policy is activated, StorageGRID places object data as follows:

- The data tiered from FabricPool in `fabricpool-bucket` will be erasure coded using the 2+1 erasure-coding scheme. Two data fragments and one parity fragment will be placed on three different Storage

Nodes.

- All objects in all other buckets will be replicated. Two copies will be created and placed on two different Storage Nodes.
- The copies will be maintained in StorageGRID forever. StorageGRID ILM won't delete these objects.

Create a traffic classification policy for FabricPool

You can optionally design a StorageGRID traffic classification policy to optimize quality of service for the FabricPool workload.

For details about this task, see Manage traffic classification policies. To use the FabricPool setup wizard to complete this task, go to Access and complete the FabricPool setup wizard.

Before you begin

- You are signed in to the Grid Manager using a supported web browser.
- You have the Root access permission.

About this task

The best practices for creating a traffic classification policy for FabricPool depend on the workload, as follows:

- If you plan to tier FabricPool primary workload data to StorageGRID, you should ensure that the FabricPool workload has most of the bandwidth. You can create a traffic classification policy to limit all other workloads.

  In general, FabricPool read operations are more important to prioritize than write operations.

  For example, if other S3 clients use this StorageGRID system, you should create a traffic classification policy. You can limit network traffic for the other buckets, tenants, IP subnets, or load balancer endpoints.

*Generally, you should not impose quality of service limits on any FabricPool workload; you should only limit the other workloads.

- The limits placed on other workloads should account for the behavior of those workloads. The limits imposed will also vary based on the sizing and capabilities of your grid and what the expected amount of utilization is.

Steps

1. Select **CONFIGURATION** > **Network** > **Traffic classification**.
2. Select **Create**.
3. Enter a name and a description (optional) for the policy and select **Continue**.
4. For the Add matching rules step, add at least one rule.
   a. Select **Add rule**
   b. For Type, select **Load balancer endpoint**, and select the load balancer endpoint you created for FabricPool.

      You can also select the FabricPool tenant account or bucket.

   c. If you want this traffic policy to limit traffic for the other endpoints, select **Inverse match**.

5. Optionally, add one or more limits to control the network traffic matched by the rule.

> StorageGRID collects metrics even if you don't add any limits, so you can understand traffic trends.

   a. Select **Add a limit**.
   b. Select the type of traffic you want to limit and the limit to apply.
6. Select **Continue**.
7. Read and review the Traffic classification policy. Use the **Previous** button to go back and make changes as required. When you are satisfied with the policy, select **Save and continue**.

After your finish

View network traffic metrics to verify that the polices are enforcing the traffic limits you expect.

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